Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/03/2005
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Project Title:	LOW VELOCITY FLOW BOILING OF ORDINARY LIQUIDS WITH EXTENSION TO LIQUID METALS- THE EFFECT OF REDUCED GRAVITY		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-8357	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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	Applications of boiling heat transfer in space can be found in the areas of thermal management, fluid handling and control, power systems, on-orbit storage and supply systems for cryogenic propellants and life support fluids, and for cooling of electronic packages for power systems associated with various instrumentation and control systems. Recent interest in exploration of Mars and other planets, and the concepts of in-situ resource utilization on Mars highlights the need to understand the effect of gravity on boiling heat transfer at gravity levels varying from 1 >= g/g_e >= 10^(-6). The objective of the proposed wok is to develop a mechanistic understanding of nucleate boiling and critical heat flux under low and micro-gravity conditions when the velocity of the imposed flow is small. For pool boiling, the effect of reduced gravity is to stretch both the length scale as well as the time scale for the boiling process. At high flow velocities, the inertia of the liquid determines the time and the length scales and as such the gravitational acceleration		

Task Description:	plays little role. However, at low velocities and at low gravity levels both liquid inertia and buoyancy are of equal importance. At present, we have little understanding of the interacting roles of gravity and liquid inertia on the nucleate boiling process. Little data that has been reported in the literature does not have much practical value in that it can not serve as a basis for design of heat exchange components to be used in space. The proposed work consists of both experimental and complete numerical simulations of the low velocity, low-gravity nucleate boiling process. A building block type of approach will be used in that first the growth and detachment process of a single bubble and flow and heat transfer associated with the sliding motion of the bubble over the heater surface after detachment will be studied. Thereafter, the complexity of the experiments with respect to the number of nucleation sites will be increased in steps. Liquid subcooling and flow velocity will be varied parametrically. The experiments are proposed to be conducted at 1 g_e, 10^(-2) g_e. The experiments at 10^(-2_ g_e will be conducted in the KC-135. In the laboratory experiments, particle velocimetry and holographic interferometry will be used to obtain data on velocity and temperature fields associated with a bubble prior to, and after detachment and during sliding motion. Numerical simulations modeling the micro and macro regions of the bubble will be carried out in three dimensions. Scaling the effect of gravity will be a prerequisite of the model. The results of the experiments will be used to validate analytical/numerical models. Task 1: Experiments at Earth Normal Gravity: The basic experiments with single, and merged bubbles attached to the heater surface and bubbles sliding along the heated wall will be carried out on surfaces inclined at different angles to the horizontal and subjected to forced flow parallel to the heater. The experiments will be conducted at one atmosphere pressure, but liquid subcooling and flow velocity will be varied parametrically. Water and PF-5060 will be used as test liquids. Task 2: Numerical Simulations/Models: In this task, transient temperature and flow field in vapor and liquid will be determined during growth of a single bubble subjected to flow parallel to the heater surface. Simulations will include the evaporation at the vapor-liquid interface and development of a microlayer underneath the bubble. This effort will be followed by simulations of situations in which neighboring bubbles merge, and bubbles slide along the heated surface for a long distance. The direction and magnitude of gravity, as well as flow velocity and subcooling will be variables of the problem. Task 3: Experiments in the KC-135 Aircraft: To understand bubble growth, detachment and subsequent motion of single and large merged bubbles, boiling experiments will be conducted under low gravity conditions (g/g_o 10^(-2)) of the aircraft. In these experiments, "designed" surfaces will be used. Visual observations and heat transfer data will be taken. The apparatus used in the laboratory experiments will also be employed with some modifications for experiments in the aircraft. The proposed research will provide fundamental understanding of bubble dynamics and heat transfer during nucleate boiling under low gravity and low velocity conditions. Such an understanding is needed for optimum design and safe operation of heat exchange equipment employing phase change for transfer of heat under microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the current year, effort has been expanded to complete all three of the identified tasks. Experiments on surfaces inclined at different angles to the gravitational vector were conducted to study, explicitly, the effect of liquid subcooling on bubble dynamics. Numerical simulation tools were expanded to 3-D and calculations have been carried out for bubble growth, bubble sliding motion prior to lift off, and bubble lift off from the heater surface. In the numerical simulations, the orientation of the gravity vector normal to the heater, flow velocity, and liquid subcooling are the key variables. The results of numerical simulations show that predicted bubble lift off diameters are generally larger than those observed in experiments. We are currently investigating the reason for the difference. A rig for bubble growth, detachment, and subsequent motion of single and large merged bubbles under the low gravity environment of the KC-135 aircraft has been developed. We began conducting flow boiling experiments in the KC-135 aircraft in the Fall of 2004.
Bibliography Type:	Description: (Last Updated: 08/23/2012)
Papers from Meeting Proceedings	Li D, Dhir VK. "A Numerical Study of Single Bubble Dynamics During Flow Boiling." ASME 2004 International Mechanical Engineering Congress and Exposition (IMECE2004), Anaheim, California, November 13 – 19, 2004. ASME 2004 International Mechanical Engineering Congress and Exposition (IMECE2004), Anaheim, California, November 13 – 19, 2004. Paper # 2004-59296, p. 485-492. http://dx.doi.org/10.1115/IMECE2004-59296 , Nov-2004
Papers from Meeting Proceedings	Manickam S, Dhir VK. "Holographic Interferometric Study of Heat Transfer Associated with a Single Vapor Bubble Sliding along a Downward Facing Heater Surface." ASME Summer Heat Transfer Conference, Las Vegas, Nevada, July 21-23, 2003. Proceedings of ASME Summer Heat Transfer Conference, Las Vegas, Nevada, July 2003. Paper no. HT2003-47159; p. 317-327. http://dx.doi.org/10.1115/HT2003-47159 , Jul-2003